APPM 4600 — HOMEWORK # 10

For all homeworks, you should use Python. **Do not use** symbolic software such as Maple or Mathematica.

1. Consider the task of approximating

$$I = \int_0^{\pi} e^x \cos(x) dx.$$

The true value of I is $I = -12.070346316389632$.

   (a) Use Tapezoidal and Simpson's rule to approximate the integral by decreasing the subinterval length h by a factor of 2 until get a relative error of $1e-10$.

   (b) On a log-log scale, plot the absolute error vs h for the two methods. What are the slope of the corresponding lines? How does this compare with the error theorems we discussed in class?

2. Consider the task of approximating the following integral $\int_{-5}^{5} \frac{1}{1+s^2} ds$ using Trapezoidal rule and Simpson's rule with equally spaced points.
For Trapezoidal rule, this means the interval $[-5, 5]$ is partitioned into equally spaced points $t_0, t_1, \ldots, t_n$.
For Simpson's rule, this means the interval $[-5, 5]$ is partitioned into equally spaced points $t_0, t_1, \ldots, t_n$ where $n = 2k$ is even. The even indexed points should be the endpoints of your subintervals.

   (a) Use the error estimates derived in class to choose n so that

$$\left| \int_{-5}^{5} \frac{1}{1+s^2} ds - T_n \right| < 10^{-4} \quad \text{and} \quad \left| \int_{-5}^{5} \frac{1}{1+s^2} ds - S_n \right| < 10^{-4},$$

   where T_n is the result of the composite Trapezoidal rule and where S_n is the result of the composite Simpson's rule. Be sure to explain your reasoning for choosing n in both cases (these n values will be different in the two cases).

   (b) Run your code with the predicted values of n and compare your computed values S_n and T_n with that of `SCIPY`'s `quad` routine on the same problem. Run the built in quadrature twice, once with the default tolerance of 10^{-6} and another time with the set tolerance of 10^{-4}. Report the number of function evaluations required in both cases and compare these to the number of function values your codes (both S_n and T_n) required to meet the tolerance

   Turn in your codes and the results of this test.

3. Find the constants x_0, x_1 and c_1 so that the quadrature formula

$$\int_0^1 f(x) dx = \frac{1}{2} f(x_0) + c_1 f(x_1)$$

has the highest possible degree of precision.

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